



Department of Electronics and Communication Engineering

Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: VII Semester B.E. ECE A

Course Code & Title: EC8701 Antennas and Microwave Engineering

Name of the Faculty member: Mr.D.Gopinath

Innovative Practice Description

- **Unit / Topic: Unit IV / Power Divider and Magic Tee**
- **Course Outcome: CO4** The students will be able to explain the basic concept of various microwave devices.
- **Topic Learning Outcome: TLO9** The students will be able to describe the basic principles of Microwave Passive components
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Power divider and the magic tee are the passive microwave devices which can be understood easily. Flipped classroom practice was chosen because the topic is easy and interesting for the students to refer by themselves.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

The students were given with the reference materials to get prepared. The students were asked to make a group among them. The questions were asked from the topic they have prepared. As it is power divider and Magic Tee, most of the questions were related to the ports. As a group, they have discussed and came up with answer for the questions. After that, they were asked to get the class back together to share the individual's group work with everyone. While discussing like this, they can get an in-depth knowledge about the topic and they can clearly express their ideologies related to that topic.

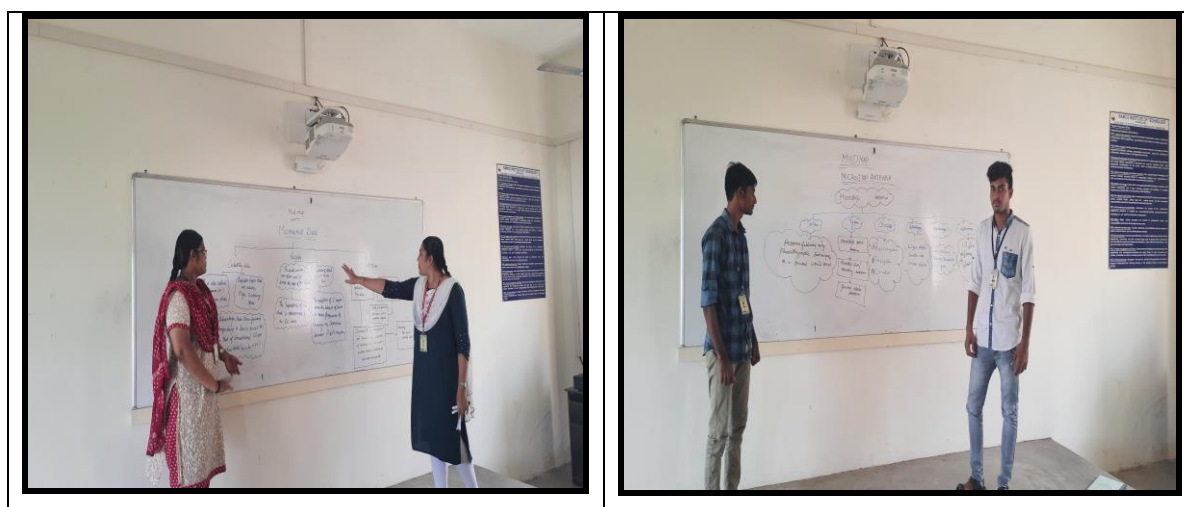
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PSO3
CO5	3	2	3	3	3	2

- **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	3	3	3	2
Justification for correlation	Electromagnetic Fields fundamentals are applied	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and development of microwave passive devices

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

1. They can get engaged in class for the entire 50 minutes.
2. The activity acts as a better platform for discussion.
3. It is quite interesting.

❖ **Benefit of the practice:**

1. They will get exposed to text books and reference materials.
2. Better interaction and communication due to group activity.
3. All the students can effectively take part in the activity and the idea about the topic has been reached more students.

❖ **Challenges faced in implementation:**

1. To organize the activity little more time has been consumed than expected.

References:

1. <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>



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Innovative Practice Description

- **Unit / Topic:** Unit V / Impedance Transformation and Impedance Matching
- **Course Outcome:** CO5 The students will be able to design a microwave system for the given application.
- **Topic Learning Outcome:** TLO12 The students will be able to describe the importance of Impedance matching and Impedance transformation
- **Activity Chosen:** Team Assisted Individualization
- **Justification:**

Impedance matching and Impedance transformation methods are quite interesting and includes more sub topics like Single Stub matching, Double stub matching and Quarter-wave Transformer. As students were already exposed to these topics in their previous semester subject Transmission lines and RF Systems. Cooperative Learning refers to learning methods in which students work together in small groups and help each other in education. Cooperative learning model is needed that improves not only social interaction but also student learning performance, one of which is to use a Cooperative Learning Model of Team Assisted Individualization (TAI).

This TAI activity can make them to remember and recall what they have already learnt.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

Initially the students were instructed to form a group among themselves. Around seven groups were formed with four in each. For the first 5 minutes, they were asked to select a name for their group which should be classy and highly correlated to subject. After selecting the name, they were asked to explain meaning of their group name and why they have chosen it. After that all the groups were assigned with some questions related to topics single stub matching, Double stub matching and Quarter Wave Transformer. Entire topics were divided as modules. They were allocated with 10 minutes for each module.

They have discussed the answers and justification among their team. After completing each module, the answers for that module were discussed in common. In between the modules, to keep them engaged, activities like class poll have been conducted.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PSO3
CO5	3	2	3	3	2	2	3	3

• **PO / PSO mapped:**

Innovative practice	PO1	PO3	PO8	PO9	PO10	PSO3
	3	3	2	2	3	3
Justification for correlation	Transmission lines fundamentals are applied	Design of impedance matching network	Group activity which requires inter-personal ethics	Needs individual and team work	Effective communication is required	Students involves design and development of stub

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

1. They can get engaged in class for the entire 50 minutes.
2. The activity acts as a better platform for discussion.
3. It is quite interesting.

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

1. They can easily understand while discussing and solving it as a group.

2. Better interaction and communication due to group activity.
3. Multiple ways to solve a problem can be obtained as all the groups are assigned with same problem statement.

Based on the class action research results, the cooperative learning model of the Team Assisted Individualization can improve social interaction and student learning achievement

❖ ***Challenges faced in implementation:***

1. To organize the activity little more time has been consumed than expected.

References:

1. Team-Assisted Individualization: A Cooperative Learning Solution for Adaptive Instruction in Mathematics by Slavin, Robert E.

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Active Learning Activity- Unit-I- Introduction to Microwave Systems and Antennas

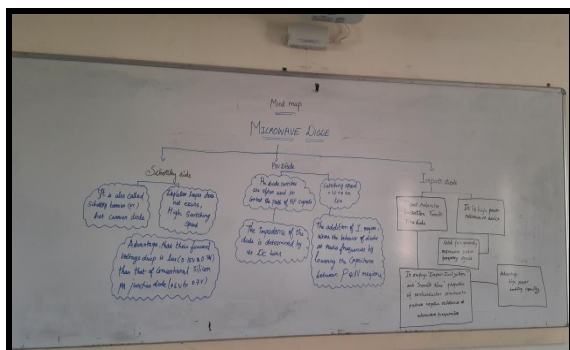
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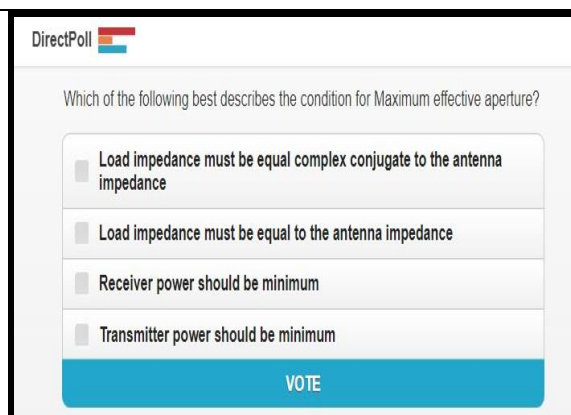
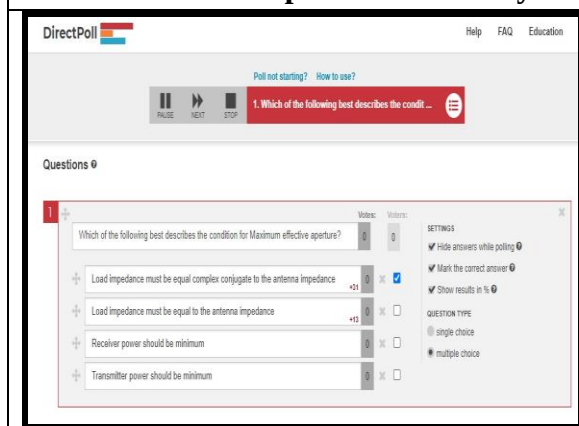
Name of the Faculty member: Mr.D.Gopinath

Name of the Topic	Name of the Activity	Date & Duration
Near- and Far-field regions	Mind map	02.09.2021&5 Minutes
Aperture Efficiency and Effective Area	Class Poll	03.09.2021 &5 Minutes
Link budget and link margin	One Minute Paper	03.09.2021 &5 Minutes

Near- and Far-field regions- Mind map



Aperture Efficiency and Effective Area- Class Poll





Link budget and link margin- One Minute Paper

ONE MINUTE PAPER 953618106003
ECE701-AME- LINK BUDGET LINK MARGIN

Path loss (dB) $L_0 = 20 \lg\left(\frac{4\pi R}{\lambda}\right) > 0$
path loss depends on wavelength

Considering,

- Transmit power (P_t)
- Transmit antenna line loss ($-L_t$)
- Transmit antenna gain (G_t)
- Path loss ($-L_0$)
- Atmospheric attenuation ($-L_A$)
- Receive antenna gain (G_r)
- Receive antenna line loss ($-L_r$)
- Receive power P_r

$P_r(\text{dBm}) = P_t - L_t + G_t - L_0 - L_A + G_r - L_r$

Active Learning Activity- Unit-II- Radiation Mechanisms and Design Aspects

Name of the Topic	Name of the Activity	Date & Duration
Aperture antennas	One Minute Paper	15.09.2021&5 Minutes
Microstrip antennas	Mind-map	21.09.2021&5 Minutes
Frequency independent antennas- Design considerations and applications	Think pair Share	22.09.2021&10 Minutes

Aperture antennas- One Minute Paper

One minute paper 953618106006
Antenna and Microwave Engineering
Aperture Antennas

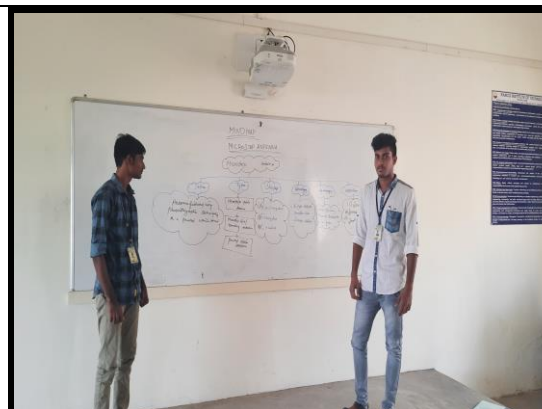
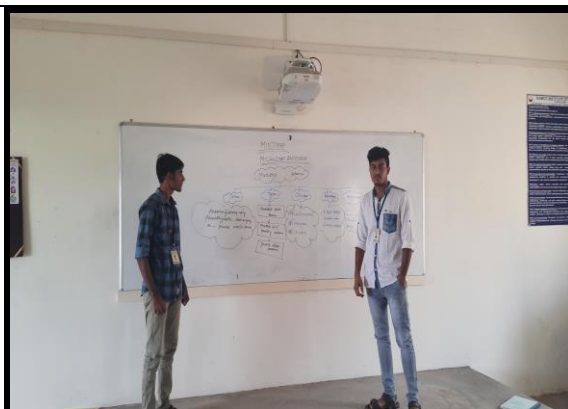
1. Reflector Antennas
2. Microstrip patch Antennas
3. Helical Antennas
4. Slot Antennas
5. Horn Antennas



Microstrip antennas- Mind-map



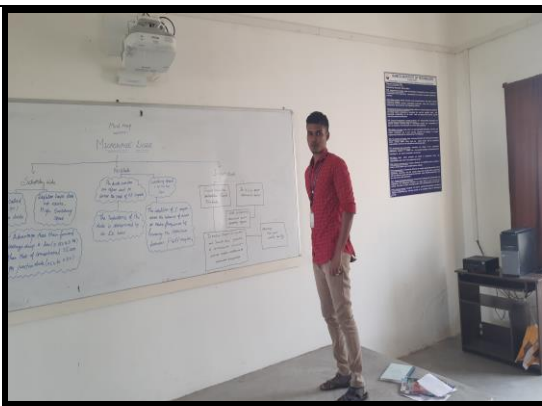
Frequency independent antennas- Design considerations and applications- Think Pair Share



Active Learning Activity- Unit-III- Antenna Arrays and Applications

Name of the Topic	Name of the Activity	Date & Duration
Introduction to Antenna Array and Applications	Brain Storming	25.09.2021&5 Minutes
Uniformly spaced arrays with non-uniform excitation amplitudes	Open Book Test	07.10.2021&10 Minutes

Introduction to Antenna Array and Applications- Brain Storming





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Uniformly spaced arrays with non-uniform excitation amplitudes- Open Book Test



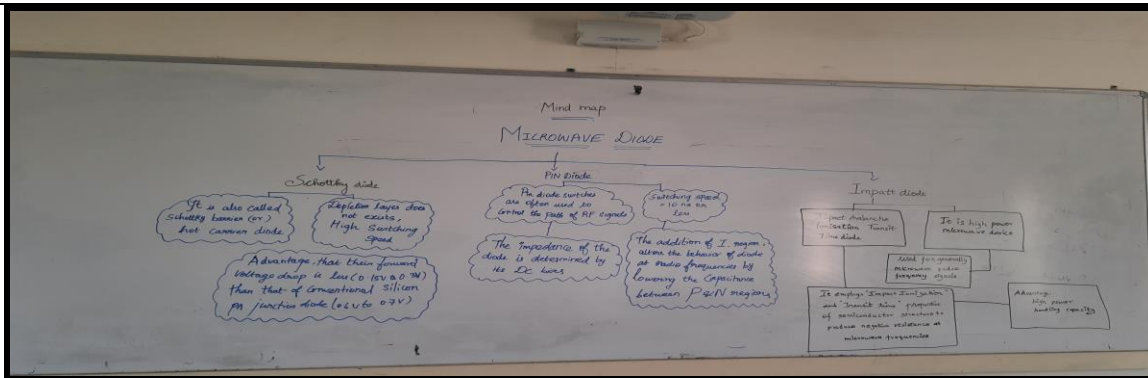
Active Learning Activity- Unit-IV- Passive and Active Microwave Devices

Name of the Topic	Name of the Activity	Date & Duration
Power Divider, Magic Tee	Hardware Demonstration	18.10.2021&10 Minutes
IMPATT diodes, Schottky Barrier diodes, PIN diodes	Mind map	23.10.2021&5 Minutes
Magnetron	Zero Minute Speech	02.11.2021&5 Minutes

Power Divider, Magic Tee- Hardware Demonstration



IMPATT diodes, Schottky Barrier diodes, PIN diodes- Mind map





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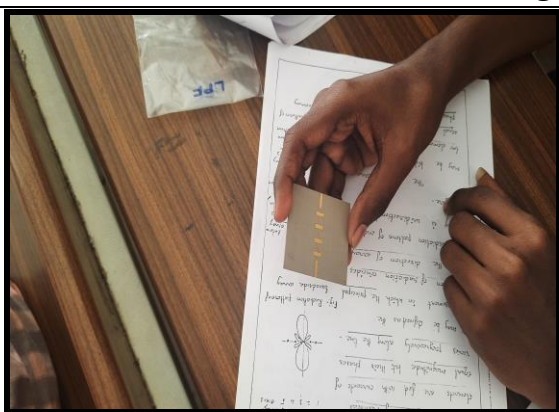
Magnetron- Zero Minute Speech



Active Learning Activity- Unit-V- Microwave Design Principles

Name of the Topic	Name of the Activity	Date & Duration
Microwave Filter Design	Hardware Demonstration	15.11.2021&10 Minutes
Low Noise Amplifier Design	Open Book Test	18.11.2021&10 Minutes
Microwave Oscillator Design	Class Poll	20.11.2021&5 Minutes

Microwave Filter Design- Hardware Demonstration



Low Noise Amplifier Design- Open Book Test





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Microwave Oscillator Design- Class Poll

DirectPoll

Any device with negative impedance as its characteristic property can be called:

- ☐ Energy source
- ☐ Energy sink
- ☐ Oscillator
- ☐ None of the mentioned

VOTE

Any device with negative impedance as its characteristic property can be called:

Energy source 0%

Energy sink 0%

Oscillator 0%

None of the mentioned 0%